

Risk factors for postoperative delirium in adult patients undergoing cardiopulmonary bypass in cardiac surgery.

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To identify risk factors for postoperative delirium in adult patients undergoing cardiopulmonary bypass. The medical records of 214 patients who underwent cardiopulmonary bypass at the First Affiliated Hospital of Hunan Normal University from January 2022 to January 2024 were retrospectively analyzed. Based on the occurrence of postoperative delirium, patients were divided into a delirium group (49 cases) and a non-delirium group (165 cases). Outcome measures, including clinical baseline data, preoperative left ventricular ejection fraction (LVEF), preoperative Acute Physiology and Chronic Health Evaluation II (APACHE II) score, anesthesia duration, surgery duration, aortic cross-clamp duration, intraoperative mean arterial pressure (MAP), intraoperative blood loss, length of intensive care unit (ICU) stay, postoperative mechanical ventilation duration, as well as postoperative MAP, were compared between the two groups. Logistic regression analysis was performed to identify the potential factors associated with post-operative delirium in patients. Of the included 214 patients, 49 patients experienced delirium, resulting in an incidence of 22.90%. The delirium group had significantly lower LVEF and significantly higher APACHE II scores compared to the non-delirium group (all $P < 0.05$). The delirium group experienced significantly longer length of ICU stay and postoperative mechanical ventilation duration as compared to the non-delirium group (all $P < 0.05$). Logistic regression analysis identified preoperative New York Heart Association classification (NYHA) (OR: 6.755, 95% CI: 2.529-18.039, $P < 0.001$), preoperative LVEF (OR: 6.886, 95% CI: 2.383-19.899, $P < 0.001$), preoperative APACHE II score (OR: 7.788, 95% CI: 2.740-22.135, $P < 0.001$), length of ICU stay (OR: 9.463, 95% CI: 3.563-25.135, $P < 0.001$), and postoperative mechanical ventilation duration (OR: 9.744, 95% CI: 3.419-27.768, $P < 0.001$) as independent factors influencing the occurrence of delirium in adult patients undergoing cardiopulmonary bypass. Postoperative delirium is a prevalent complication among patients undergoing cardiac surgery. Independent risk factors for postoperative delirium in adult patients undergoing cardiopulmonary bypass during cardiac surgery include preoperative NYHA class III or IV, lower LVEF, higher APACHE II score, an extended length of stay in the ICU, and prolonged postoperative mechanical ventilation duration.

Clinical characteristics of delirium in older patients with first-ever acute myocardial infarction who underwent percutaneous coronary intervention : A retrospective study.

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Delirium is a serious complication of cardiac surgery and a common clinical problem. The study aimed to identify the incidence, risk factors, and outcomes of delirium in older patients (≥ 65 years) with first-ever acute myocardial infarction (AMI) who underwent percutaneous coronary intervention (PCI). A retrospective cohort study was performed in a hospital in northern China. A total of 1033 older patients with first-ever AMI who underwent PCI between January 2018 and April 2021 were screened for delirium using the CAM-ICU method. Clinical and laboratory data were collected. A total of 134 (12.97%) patients were diagnosed with delirium. Patients with delirium were older. The most common concomitant diseases were cardiac arrest, chronic renal failure, and a history of coronary artery bypass graft (CABG). Delirious patients experienced more times of mechanical ventilation, more intra-aortic balloon pump (IABP) support, high postoperative immediate pain score (VAS), more non-bedside cardiac rehabilitation, and longer total length of stay and cardiac care unit (CCU) time. Multivariable logistic regression showed that age, mechanical ventilation, postoperative immediate pain score, and non-bedside cardiac rehabilitation

were independently associated with delirium. Delirium was an independent predictor of prolonged CCU stay, total length of stay, and 1-year mortality. Age, mechanical ventilation, postoperative immediate pain score, and non-bedside cardiac rehabilitation were independently closely related to delirium in older patients with first-ever AMI who underwent PCI. Delirium was associated with a higher 1-year all-cause mortality.

Delirium after coronary artery bypass graft surgery and late mortality.

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Delirium is common after cardiac surgery, although under-recognized, and its long-term consequences are likely underestimated. The primary goal of this study was to determine whether patients with delirium after coronary artery bypass graft (CABG) surgery have higher long-term out-of-hospital mortality when compared with CABG patients without delirium. We studied 5,034 consecutive patients undergoing CABG surgery at a single institution from 1997 to 2007. Presence or absence of neurologic complications, including delirium, was assessed prospectively. Survival analysis was performed to determine the role of delirium in the hazard of death, including a propensity score to adjust for potential confounders. These analyses were repeated to determine the association between postoperative stroke and long-term mortality. Individuals with delirium had an increased hazard of death (adjusted hazard ratio [HR], 1.65; 95% confidence interval [CI], 1.38-1.97) up to 10 years postoperatively, after adjustment for perioperative and vascular risk factors. Patients with postoperative stroke had a HR of 2.34 (95% CI, 1.87-2.92). The effect of delirium on subsequent mortality was the strongest among those without a prior stroke (HR 1.83 vs HR 1.11 [with a prior stroke] [p-interaction = 0.02]) or who were younger (HR 2.42 [≥65 years old] [p-interaction = 0.04]). Delirium after cardiac surgery is a strong independent predictor of mortality up to 10 years postoperatively, especially in younger individuals and in those without prior stroke. Future studies are needed to determine the impact of delirium prevention and/or treatment in long-term patient mortality.

Delirium: a cause for concern beyond the immediate postoperative period.

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Delirium is a common neurologic complication after cardiac surgery, and may be associated with increased morbidity and mortality. Research has focused on potential causes of delirium, with little attention to its sequelae. Perioperative data were collected prospectively on all isolated cases of coronary artery bypass grafting (CABG) performed from 1995 to 2006 at a single center. The definition of delirium used in the study was that of the Society of Thoracic Surgeons. Characteristics of patients who became delirious postoperatively were compared with those of patients who did not. The outcomes of interest were long-term all-cause mortality, hospital admission for stroke, and in-hospital mortality, examined in all three cases through multivariate analysis. Of 8,474 patients who underwent CABG within the defined period, 496 (5.8%) developed postoperative delirium and 229 (2.7%) died while in the hospital. At baseline, patients who developed delirium were more likely to be older and to have a greater burden of comorbid illness. Delirium was an independent predictor of perioperative stroke (odds ratio [OR]; 1.96; 95% confidence interval [CI], 1.22 to 3.16), but was not associated with in-hospital mortality (OR, 0.81; 95%CI, 0.49 to 1.34). Delirious patients had a median postoperative hospital stay of 12 days (interquartile range [IQR], 8 to 21 days) versus 6 days (IQR, 5 to 8 days) for those who were nondelirious.

Delirium was identified as an independent predictor of all-cause mortality (hazard ratio [HR], 1.52; 95%CI, 1.29 to 1.78) and hospitalization for stroke (HR, 1.54; 95%CI, 1.10 to 2.17). There was an association between delirium and adverse outcomes after CABG that persisted beyond the immediate perioperative period. Patients with delirium after CABG appear to have an increased long-term risk of death and stroke. The advancing age and rising rates of delirium in the CABG population make it necessary to address the prevention and management of delirium in this population.

Perioperative blood pressure variability as a risk factor for postoperative delirium in the patients receiving cardiac surgery.

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Delirium is one of the most common neurological complications after cardiac surgery. The purpose of our study was to assess the relationship between perioperative blood pressure variability (BPV) and postoperative delirium (POD) in the patients undergoing cardiac surgery. Adult patients received cardiac surgery and stayed in Cardiovascular Intensive Care Unit (ICU) for more than 24 h after surgery during the study period between June 2019 and December 2022 were included in this study. Baseline characteristics, perioperative hemodynamic variables and postoperative laboratory results of the cardiac patients were collected and analyzed. Perioperative BPV was quantified by calculating the standard deviation (SD) and average real variability (ARV) of blood pressure. Assessment of delirium was based on the mental status of the patients and CAM-positive. The relationship between perioperative BPV and POD was analyzed by LASSO and logistic regression using R (R package, 4.3.2). The incidence of POD was 15.0% (324/2164) in the patients receiving cardiac surgery, and the average day for POD occurred at day 3 after surgery. Patients with POD had statistically lower levels of intraoperative mean blood pressure ($P = 0.015$) and blood pressure ARV ($P < 0.001$) as well as mean blood pressure at 24 h postoperatively ($P = 0.003$) when compared to those without. Whereas, ARV for systolic blood pressure (8.64 vs. 7.91 mmHg, $P < 0.001$), diastolic blood pressure (4.00 vs. 3.77 mmHg, $P = 0.014$) and mean blood pressure (5.23 vs. 4.94 mmHg, $P = 0.001$) at 24 h postoperatively was significantly higher in the patients with POD than those without. LASSO regression and further logistic regression revealed that intraoperative blood pressure ARV (OR:0.92, $P < 0.001$), mean central venous pressure (OR:1.05, $P = 0.048$) and ARV of systolic blood pressure (OR:1.17, $P = 0.002$) at 24 h postoperatively were independent risk factors for POD. Perioperative ARV, especially postoperative high ARV exposure, was associated with POD in the patients receiving cardiac surgery. Maintaining a relatively stable blood pressure after cardiac surgery might be beneficial to avoid POD in those patients.

Effect of mannitol on postoperative delirium in patients undergoing coronary artery bypass graft: a randomised controlled trial.

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Postoperative delirium (POD), especially after cardiac surgery with cardiopulmonary bypass (CPB), is a relatively common and severe complication increasing side effects, length of hospital stay, mortality and healthcare resource costs. This study aimed to determine the impact of using mannitol serum in the prime of CPB for preventing the occurrence of delirium in patients undergoing coronary artery bypass surgery. This study is a single-centre, double-blinded, randomised, controlled trial that was conducted

from December 2022 to May 2023. Patients in the age range of 18-70 who underwent elective coronary artery bypass surgery were included in the study. In the control group (n=45), the prime solution included Ringer's lactate serum. In the intervention group (n=45), the prime solution consisted of 200 mL mannitol serum and Ringer's lactate serum. The primary outcome of the study was the incidence of POD. Secondary outcomes included the duration of mechanical ventilation, length of stay in the intensive care unit (ICU) and 30-day in-hospital mortality. There were no statistically significant differences in demographic characteristics and risk factors between the control and intervention groups ($p < 0.05$). However, the incidence of POD was significantly lower in the intervention group compared with the control group (22.25% vs 42.2%, $p = 0.035$). There were no significant differences between the two groups regarding CPB time, aortic cross-clamp time, duration of mechanical ventilation and length of stay in ICU ($p < 0.05$). Additionally, mortality rates and rates of return to the operating room did not differ significantly between the two groups ($p < 0.05$). This study concluded that adding mannitol to the prime of CPB pump can help reduce the incidence of delirium after cardiac surgery. IRCT20221129056660N1.

Intranasal administration of insulin on the incidence of postoperative delirium in middle-aged patients undergoing elective on-pump cardiac surgery (INIPOD-MOPS): a prospective double-blinded randomized control study protocol.

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Delirium, marked by acute cognitive decline, poses a life-threatening issue among older individuals, especially after cardiac surgery, with prevalence ranging from 15 to 80%. Postoperative delirium is linked to increased morbidity and mortality. Although clinical trials suggest preventability, there is limited research on intranasal insulin (INI) for cardiac surgery-related delirium. INI has shown promise in managing cognitive disorders. It rapidly elevates brain hormone levels, enhancing memory even in non-impaired individuals. While effective in preventing delirium in gastrointestinal surgery, its impact after cardiac surgery remains understudied, especially for middle-aged patients. This is a prospective randomized, double-blind, single-center controlled trial. A total of 76 eligible participants scheduled for elective on-pump cardiac surgery will be enrolled and randomly assigned in a 1:1 ratio to either receive Intranasally administered insulin (INI) or intranasally administered normal saline. The primary outcome of our study is the incidence of postoperative delirium (POD). Secondary outcomes include duration of ICU, postoperative hospital length of stay, all in-hospital mortality, the change in MMSE scores pre- and post-operation, and incidence of postoperative cognitive dysfunction at 1 month, 3 months, and 6 months after operation. Moreover, we will subjectively and objectively evaluate perioperative sleep quality to investigate the potential impact of nasal insulin on the development of delirium by influencing sleep regulation. Our study will aim to assess the impact of intranasal administration of insulin on the incidence of postoperative delirium in middle-aged patients undergoing on-pump elective cardiac surgery. If intranasal insulin proves to be more effective, it may be considered as a viable alternative for preventing postoperative delirium. ChiCTR ChiCTR2400081444. Registered on March 1, 2024.

A nomogram for predicting the risk of postoperative delirium in individuals undergoing cardiovascular surgery.

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Delirium is a common mental disorder after adult cardiovascular surgery. Fifteen to 23% of patients undergoing cardiovascular surgery and cardiomyopathy experience delirium, and the efficacy of treatment interventions for delirium has been consistently unsatisfactory. A total of 729 patients who underwent cardiovascular surgery were randomly allocated into a training set and a validation set. A nomogram was developed using a logistic regression model to predict the incidence of delirium following cardiovascular surgery. The validity of the model was assessed by determining the receiver operating characteristic (ROC) curve, calculating the area under the ROC curve (AUROC), performing a calibration plot, and executing a decision curve analysis. This model was internally validated using the bootstrap method. Postoperative delirium (POD) occurred in 165 cases (22.6%) among the 729 patients. Predictors included age, transient ischemic attack, length of preoperative stay, preoperative left ventricular injection fraction and N-terminal pro-B-type natriuretic peptide level, and intraoperative infusion of dexmedetomidine and human fibrinogen. The nomogram showed sufficient differentiation and calibration (AUROC = 0.754, 95% confidence interval = 0.703-0.804). The calibration graphs showed that the predictive values of the nomogram were in agreement with the actual values. The analysis of the training and validation sets suggested that the model possessed specific clinical significance. In summary, the predictive model consists of seven factors that can roughly predict the occurrence of POD in patients who undergo cardiovascular surgery.

Diabetes and elevated preoperative HbA1c level as risk factors for postoperative delirium after cardiac surgery: an observational cohort study.

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Postoperative delirium (POD) is a common complication of cardiac surgery associated with increased mortality, morbidity, and long-term cognitive dysfunction. Diabetic patients, especially those with poor diabetes control and long-standing hyperglycemia, may be at risk of developing delirium. The aim of this study was to analyze whether the occurrence of POD in cardiac surgery is associated with diabetes or elevated preoperative glycated hemoglobin (HbA1c) level. We performed a cohort analysis of prospectively collected data from a register of cardiac surgery department of a university hospital. Delirium assessment was performed twice a day during the first 5 days after the operation based on Diagnostic Statistical Manual of Mental Disorders, fifth edition criteria. We analyzed a cohort of 3,178 consecutive patients, out of which 1,010 (31.8%) were diabetic and 502 (15.8%) were diagnosed with POD. Patients with delirium were more often diabetic (42.03% vs 29.86%, $P < 0.001$) and on oral diabetic medications (34.66% vs 24.07%, $P < 0.001$), no difference was found in patients with insulin treatment. Preoperative HbA1c was elevated above normal ($\geq 6\%$) in more delirious than nondelirious patients (44.54% vs 33.04%, $P < 0.001$), but significance was reached only in nondiabetic patients (20.44% vs 14.86%, $P = 0.018$). In univariate analysis, the diagnosis of diabetes was associated with an increased risk of developing POD (OR: 1.703, 95% CI: 1.401-2.071, $P < 0.001$), but only for patients on oral diabetic medications (OR: 1.617, 95% CI: 1.319-1.983, $P < 0.001$) and an association was noted between HbA1c and POD (OR: 1.269, 95% CI: 1.161-1.387, $P < 0.001$). Multivariate analysis controlled for diabetes showed that POD was associated with age, heart failure, preoperative creatinine, extracardiac arteriopathy, and preoperative HbA1c level. More diabetic patients develop POD after cardiac surgery than nondiabetic patients. Elevated preoperative HbA1c level is a risk factor for postcardiac surgery delirium regardless of the diagnosis of diabetes.

Risk factors of delirium after cardiac surgery: a systematic review.

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Delirium or acute confusion is a temporary mental disorder that occurs frequently among hospitalized elderly patients, but also in younger patients a delirium can develop. Patients who undergo cardiac surgery have an increased risk of developing delirium that is associated with many negative consequences. Therefore, prevention of delirium is essential. Despite the high incidence of delirium, a paucity of data on risk factors for delirium exists. The aim of this study was to summarize the available information concerning these risk factors. A literature research was performed using the PubMed, Cinahl, and Cochrane Library databases and was limited to the last 10 years. Our review revealed 27 risk factors; 12 predisposing and 15 precipitating factors for delirium after cardiac surgery. The most established predisposing risk factors were atrial fibrillation, cognitive impairment, depression, history of stroke, older age, and peripheral vascular disease. The most established precipitating risk factor was a red blood cell transfusion. An abnormal albumin level was reported as the most established precipitating risk factor among blood values tested. A low cardiac output and the use of an Intra Aortic Balloon Pump or inotropic medication seem to be the most relevant risk factors associated with a postoperative delirium. A multifactorial risk model should be applied to identify patients at an increased risk of developing delirium following elective cardiac surgery. In these patients, if possible, preventative interventions can be taken and early recognition of delirium can be realized. This could potentially decrease the incidence of delirium and negative consequences caused by a postoperative delirium.
